Secure Document Access in Power BI – Design Proposal

Introduction

In many analytics platforms, users often need to access supporting documents such as PDFs along with dashboards. Providing this within Power BI can improve usability, as users can directly view related files while analyzing data.

However, this introduces security concerns. Documents may contain sensitive information, so access needs to be controlled carefully. A secure approach is required to ensure that only authorized users can view these documents

Problem Statement

Power BI does not provide a direct and secure way to display or access documents like PDFs. While links can be added, using static or public URLs can expose files to unauthorized users.

Additionally, Power BI mainly controls access at the report level, not at the document level. This makes it difficult to manage who can view specific files.

Because of this, a solution is needed that allows document access while ensuring proper authentication, authorization, and data security.

Proposed Solution Overview

To solve this problem, documents should not be directly embedded inside Power BI. Instead, they can be stored in a secure storage system such as Azure Blob Storage or AWS S3.

When a user wants to access a document, the request can be handled through a backend service. This service verifies the user's identity and permissions and then generates a temporary secure link to the document.

These links are time-limited and cannot be reused after expiration, which helps prevent unauthorized access. This approach ensures that documents remain protected while still being accessible from the Power BI dashboard.

System Architecture

The proposed solution consists of four main components: Power BI dashboard, backend service, secure storage, and an authentication system.

The Power BI dashboard provides a link or button for users to access documents. When a user clicks on it, the request is sent to a backend API instead of directly opening the file.

The backend service checks the user's identity and verifies whether the user has permission to access the requested document. Once validated, the backend generates a temporary secure link (token-based URL) for the document stored in secure storage.

This link is time-limited and allows the user to view the document safely. After expiration, the link becomes invalid, ensuring that access remains controlled.

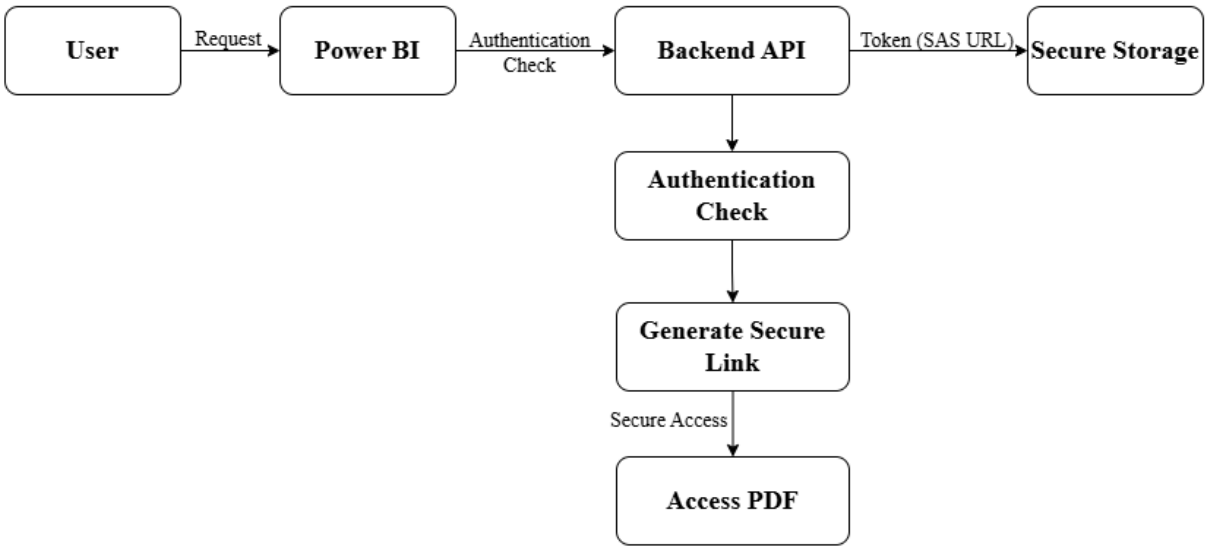

Figure: Proposed architecture for secure document access in Power BI

Security Mechanisms

To ensure secure access to documents, multiple layers of security can be applied in the proposed solution.

First, role-based access control (RBAC) can be used to ensure that users can only access documents relevant to their role or permissions. This prevents unauthorized users from viewing sensitive files.

Second, authentication can be handled using secure methods such as Azure Active Directory or token-based authentication (JWT). This ensures that only verified users can request access.

Another important mechanism is the use of time-limited secure links (such as signed URLs or SAS tokens). These links expire after a short duration, reducing the risk of misuse or sharing.

Additionally, all communication should be encrypted using HTTPS to protect data during transfer. Logging and monitoring can also be implemented to track access and detect any unusual activity.

Challenges

There are a few challenges in implementing this approach. Power BI does not support secure document embedding directly, so an external solution is required. Managing access at the document level can also be complex, especially when different users have different permissions.

Another challenge is handling expired links, as users may need to request access again. There can also be a slight delay when generating secure links through the backend service.

Trade-offs

The proposed solution improves security, but it also introduces some additional complexity. A backend service is required to handle authentication and link generation.

There may be a small increase in response time due to the extra step of validating users and generating secure links. However, this trade-off is acceptable in order to ensure proper security and controlled access.

Conclusion

The proposed solution provides a secure and practical way to enable document access within Power BI. By using a backend service and time-limited secure links, it ensures that only authorized users can access specific documents.

This approach balances usability and security, making it suitable for real-world applications where sensitive data needs to be protected. It can also be scaled easily for larger systems with multiple users and documents.